

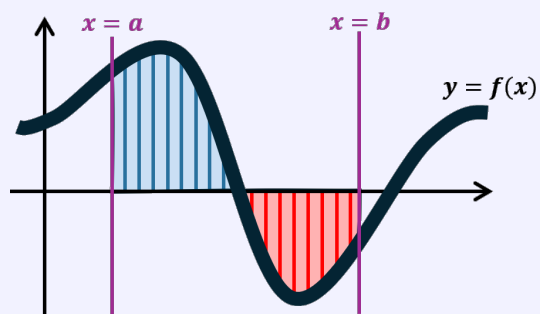
Definite Integrals

Definite Integrals

$\int_a^b f(x) dx$ computes the sum of “slices of $f(x)$ ”
from $x = a$ to $x = b$.

We understand this in terms of (signed) area of region

$$\begin{cases} x = a & \text{on the left and right sides} \\ x = b & \\ y = f(x) & \text{on the top and bottom} \\ y = 0 & \end{cases}$$



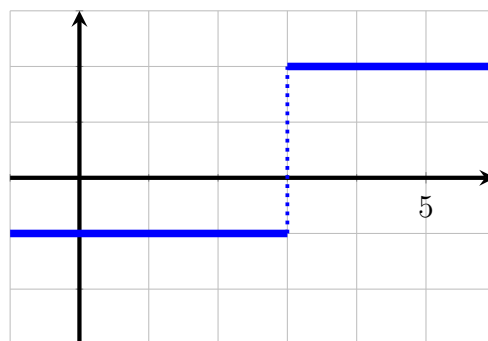
1. Compute the following definite integrals for the function graphed to the right

(A) $\int_0^3 f(x) dx$

(B) $\int_3^5 f(x) dx$

(C) $\int_0^5 f(x) dx$

(D) $\int_0^5 |f(x)| dx$



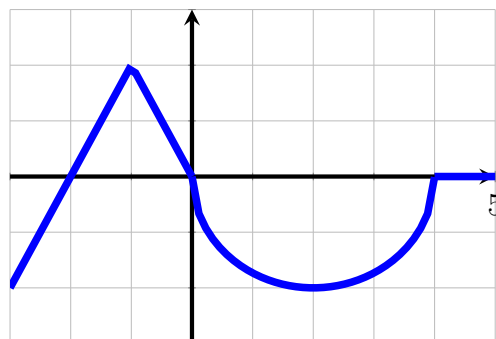
2. Compute the following definite integrals for the function graphed to the right

(A) $\int_{-2}^0 f(x) dx$

(B) $\int_0^{-2} f(x) dx$

(C) $\int_0^5 f(x) dx$

(D) $\int_{-1}^2 f(x) dx$



3. Identify the shape of the areas described and use this to compute the definite integrals below.

(A) $\int_0^5 6 \, dx$

(B) $\int_1^5 6 \, dx$

(C) $\int_0^5 6x \, dx$

(D) $\int_1^5 6x \, dx$

4. Compute the integrals below if $\int_0^1 f(x) \, dx = 2$ and $\int_1^3 f(x) \, dx = 3$

(A) $\int_0^3 f(x) \, dx$

(B) $\int_0^3 2 f(x) \, dx$

5. Compute the integrals below if $\int_0^2 f(x) \, dx = 2$ and $\int_0^4 f(x) \, dx = 8$

(A) $\int_2^0 f(x) \, dx$

(B) $\int_2^4 f(x) \, dx$